

### 1.SVM- Result

| <i>S.No</i> | <i>Hyper parameter</i> | <i>Linear (r value)</i> | <i>RBF(non-linear) r value</i> | <i>Poly(r value)</i> | <i>sigmoid(r value)</i> |
|-------------|------------------------|-------------------------|--------------------------------|----------------------|-------------------------|
| 1           | c10                    | -0.106884               | -0.1252                        | -0.1215              | -0.1228                 |
| 2           | c100                   | 0.043028                | -0.12082                       | -0.084               | -0.0964                 |
| 3           | c500                   | 0.5352                  | -0.1014                        | 0.07181              | 0.00646                 |
| 4           | c1000                  | 0.7942                  | -0.08004                       | 0.2112               | 0.13765                 |
| 5           | c2000                  | 0.87025                 | -0.0305                        | 0.49208              | 0.3607                  |
| 6           | c3000                  | 0.858411                | 0.01926                        | 0.6603               | 0.4986                  |

### 2.Decision Tree-Result

| <i>S.No</i> | <i>criterion</i>      | <i>splitter</i> | <i>R2 value</i> |
|-------------|-----------------------|-----------------|-----------------|
| 1           | friedman_mse          | best            | 0.879           |
| 2           | friedman_mse          | random          | 0.734           |
| 3           | squared_error         | random          | 0.8578          |
| 4           | squared_error         | best            | 0.8792          |
| 5           | <i>absolute_error</i> | best            | 0.893           |
| 6           | <i>absolute_error</i> | random          | 0.77            |
| 7           | poisson               | random          | 0.8468          |
| 8           | poisson               | best            | 0.911           |

### 3.RF- Final result

| <i>S.No</i> | <i>criterion</i> | <i>n_estimator</i> | <i>R2 value</i> |
|-------------|------------------|--------------------|-----------------|
| 1           | squared_error    | 50                 | 0.913731        |
| 2           | absolute_error   | 50                 | 0.9231          |
| 3           | friedman_mse     | 50                 | 0.91486         |
| 4           | poisson          | 50                 | 0.9054          |